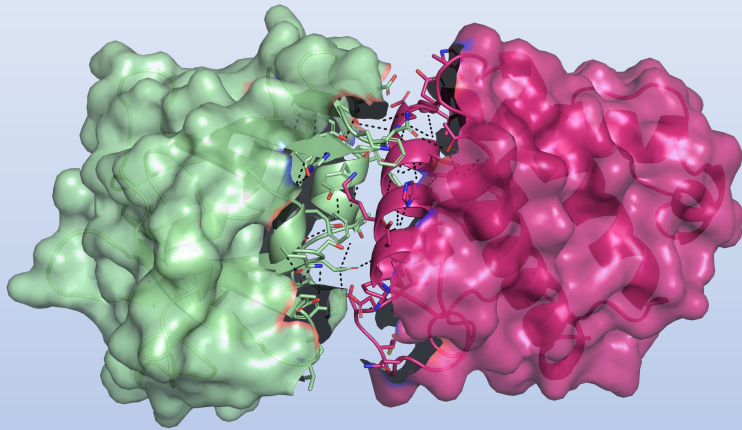


Day4: Protein Structures

Visualization of Protein Interactions



Prof Tiina A. Salminen



Structural Bioinformatics Laboratory

InFLAMES Research Flagship Center

Biochemistry, Faculty of Science and Engineering

Pre-assignments:

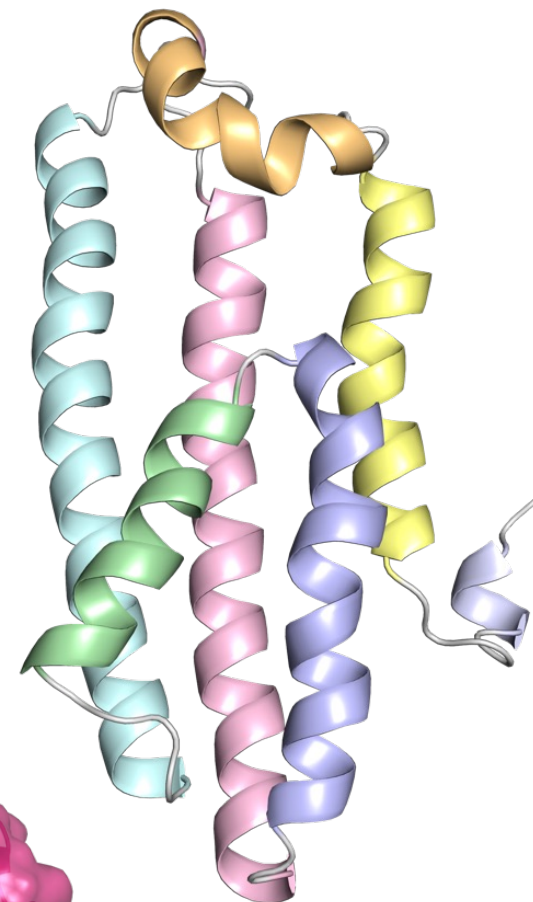
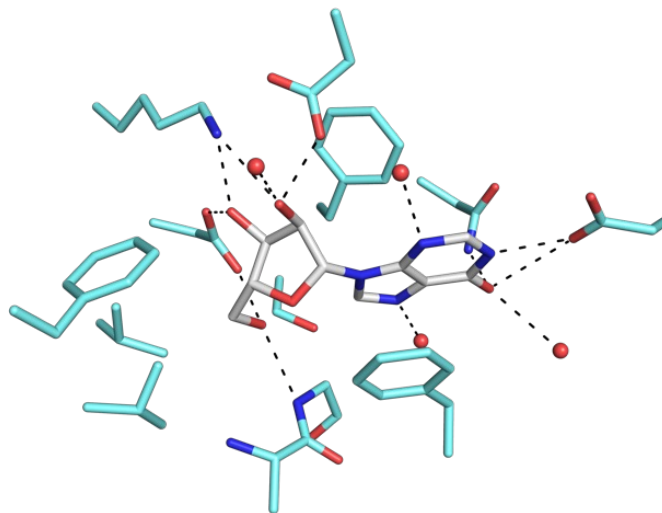
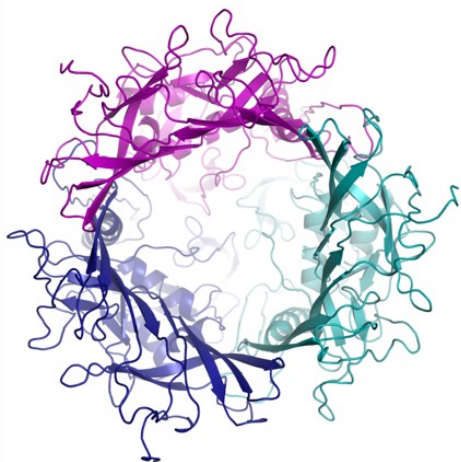
1. Install Pymol
2. Protein interactions
3. Biomolecular structures
4. AlphaFold for predicting 3D structures for proteins

On-site:

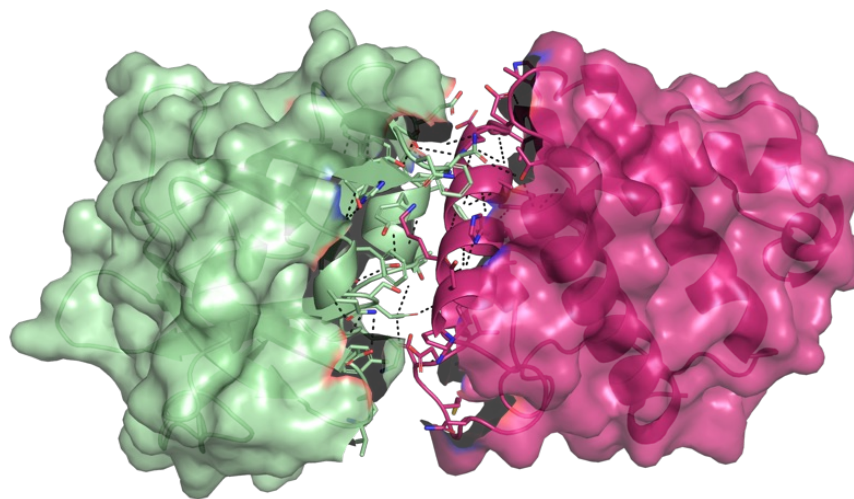
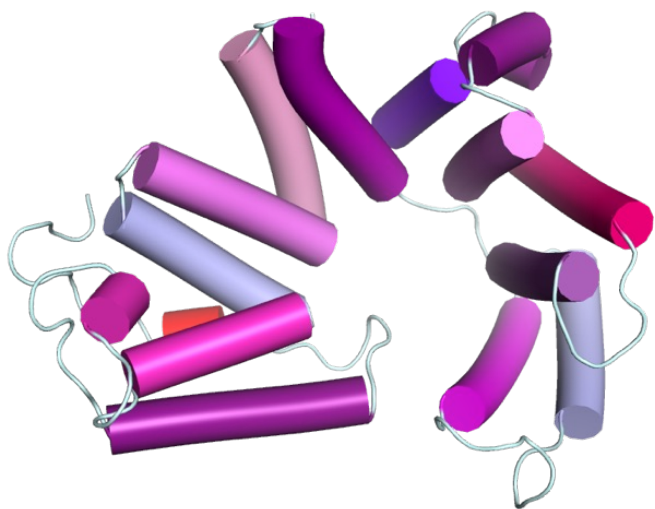
Lecture: Introduction to visualization

Exercises:

1. Individual exercise -> Learn how to use Pymol
2. Group exercise -> Presentation and project report



Visualization of 3D structures

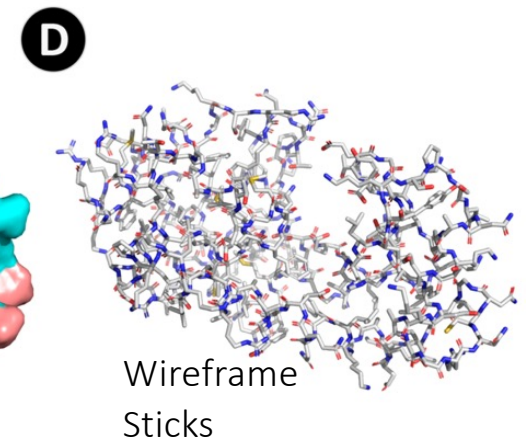
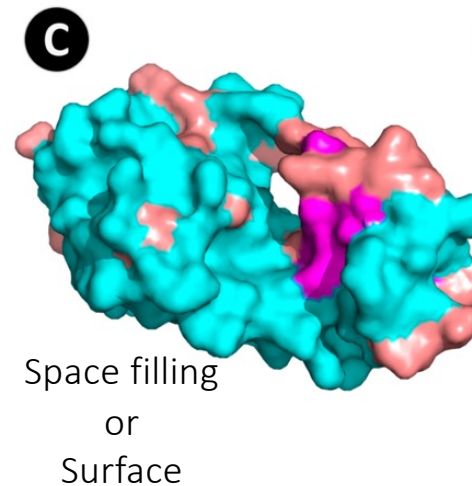
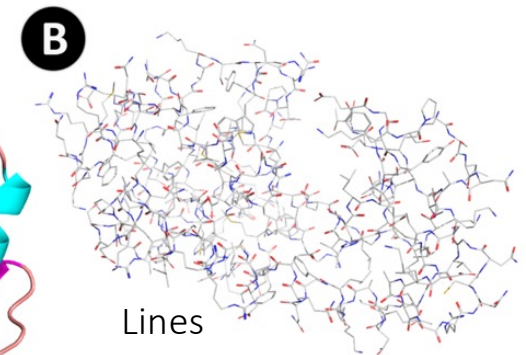
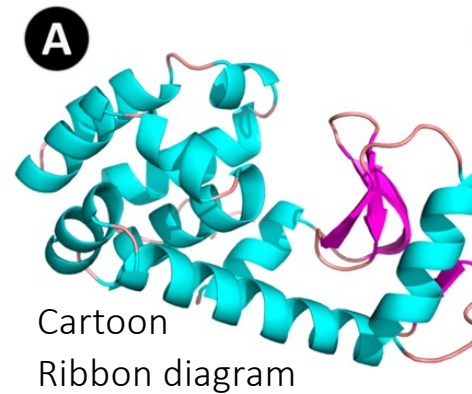


Why is visualization important?

We depend on visual information!

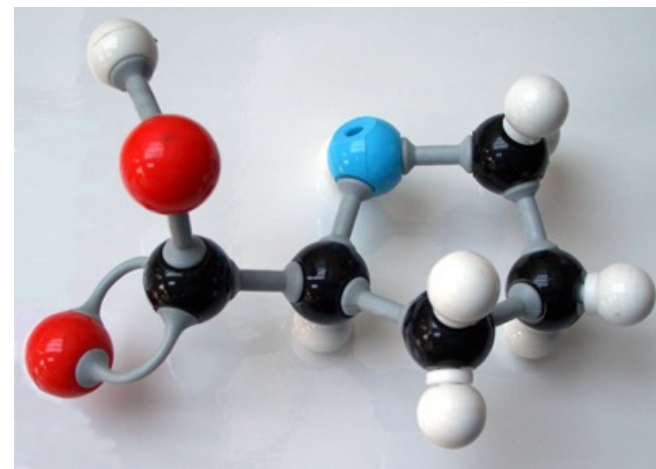
- Visualization programs enable us to:
 - **Analyze** 3D structures
 - **Study** protein interactions
 - **Understand** protein function
 - **Create** illustrative figures

Visualization styles

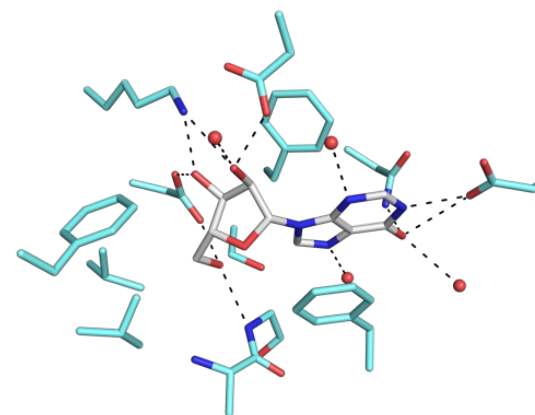


Visualization styles: Colouring

- Typical colour convention for the most common atoms
 - **Black** for carbon – any color to show details
 - **Blue** for nitrogen
 - **Red** for oxygen
- White for hydrogen
- Yellow for sulfur
- Purple for phosphorus
- Green for the halogens (F, Cl, Br, I)
- Silver for metals (Co, Fe, Ni, Cu)



A plastic ball-and-stick model of proline



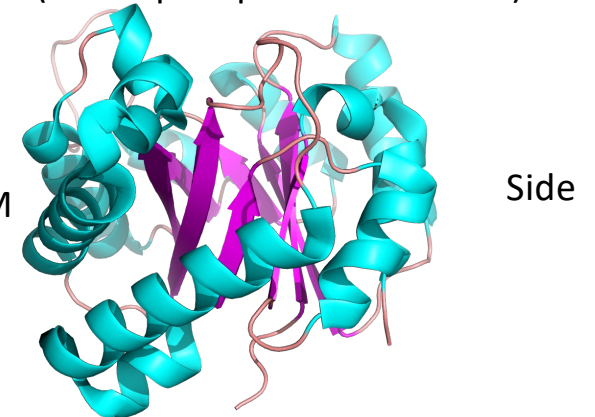
Carbon colors:
Adenosine (white) binding to BmpD protein (cyan)

Visualization styles in Pymol: Cartoon or ribbon

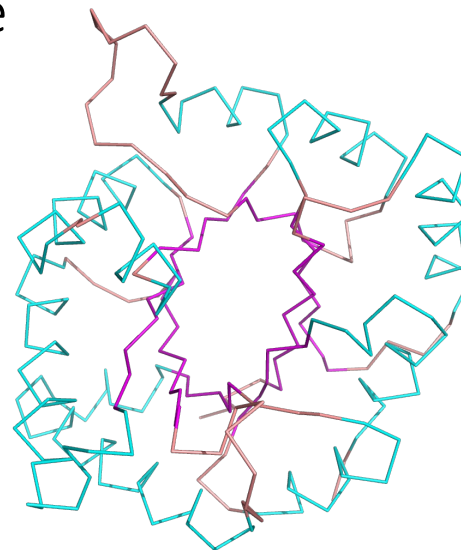
- Highlights secondary structures and overall protein fold
- **Cartoon:** shows secondary structures as arrows (β -stands) and helices (α -helices)
- **Ribbon:** CA-trace, which connects the C α atom on each amino acid to the next, revealing the overall fold

TIM barrel (triose-phosphate isomerase)

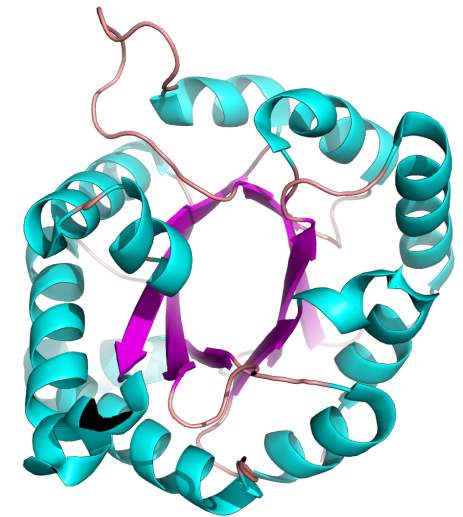
PDB ID 1TIM
Cartoon



PDB ID 1TIM
Ribbon

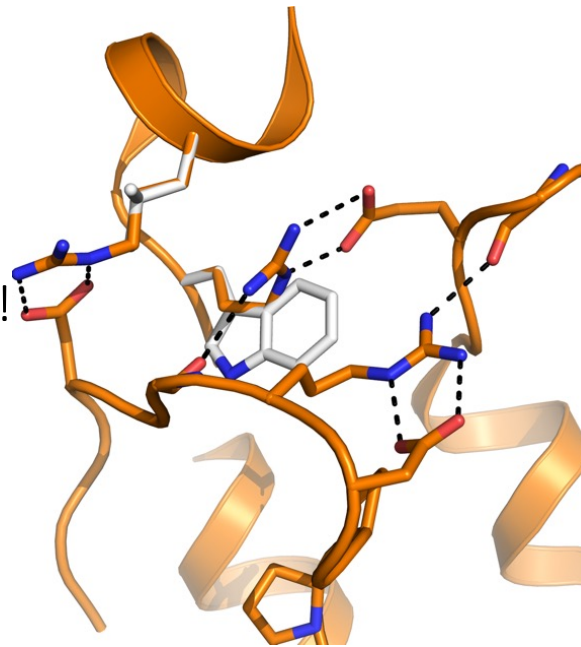


Top

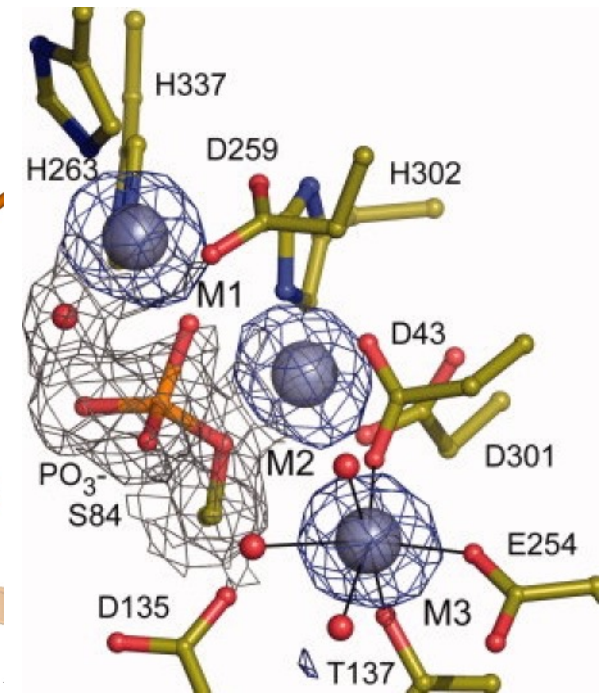


Visualization styles for atoms

- Lines, sticks and ball-and-sticks for displaying details:
 - Amino acids
 - Bonds (interactions)
 - All atoms shown in the same size!
- Waters and other atoms as spheres
 - Relative to their atom size
- Waters are often shown as red non-bonded spheres (`nb_spheres`)



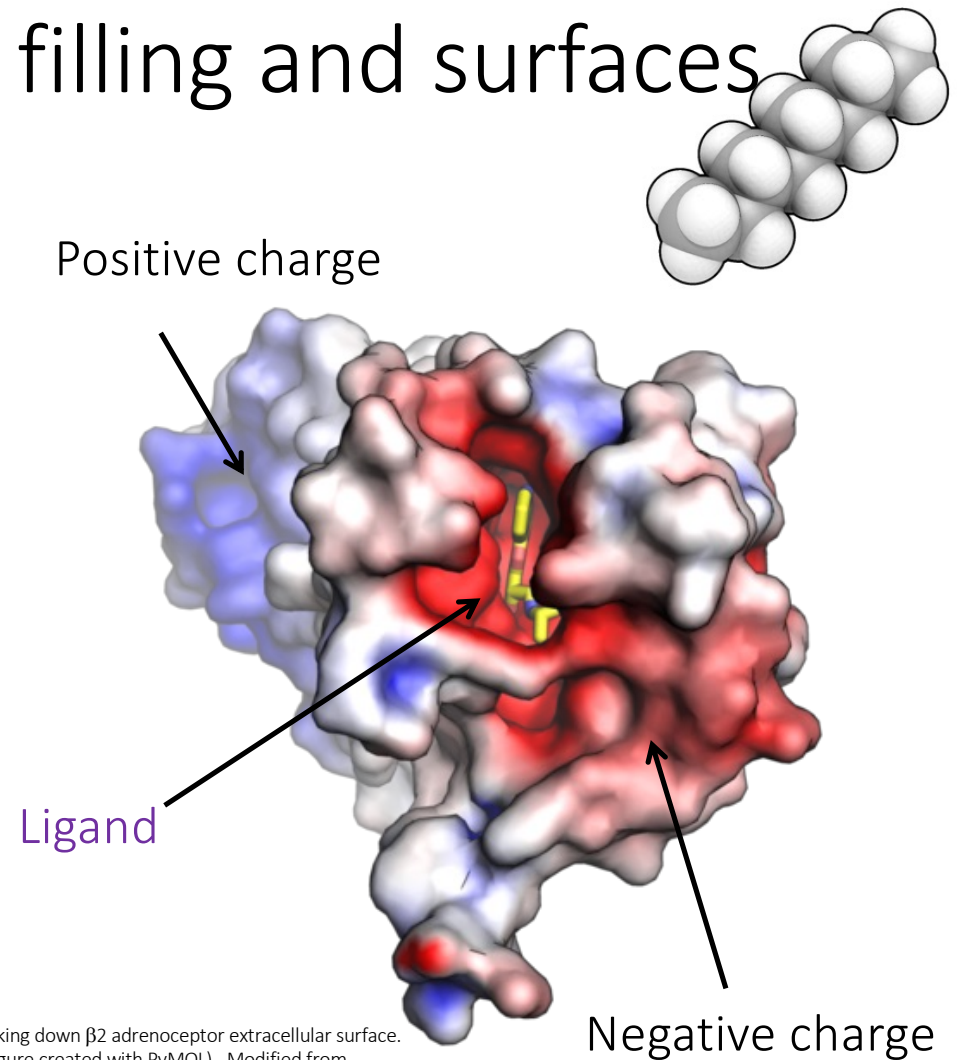
Inherited α -mannosidase mutations. Change of amino acid from R to W destroys interactions in human lysosomal α -mannosidase. Heikinheimo, unpublished



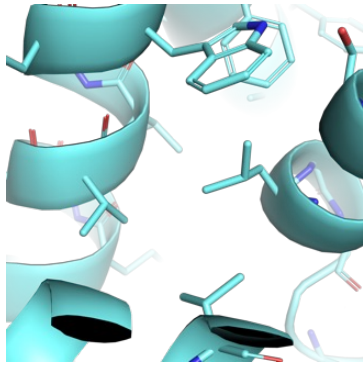
Zn^{2+} coordination and electron density map at alkaline phosphatase active site Koutsoliis 2010, doi: 10.1002/pro.284.

Visualization styles: Space filling and surfaces

- In space-filling representation, atoms are shown as spheres
 - Radius is proportional to atom radius
- Surfaces
 - Surface charges
 - Good for displaying volume, cavities, shape etc.



Charge surface looking down $\beta 2$ adrenoceptor extracellular surface.
(PDB code 2RH1, figure created with PyMOL) . Modified from
Opabinia regalis, Wikimedia commons, cc BY-SA 3.0



Hydrophobic interactions

Molecular interactions

1. Hydrophobic interactions

- Non-polar, hydrophobic residues pack together because they cannot participate in hydrogen bonding with water
 - Inside protein
 - Protein-protein interfaces

2. Polar interactions

• Hydrogen bonds

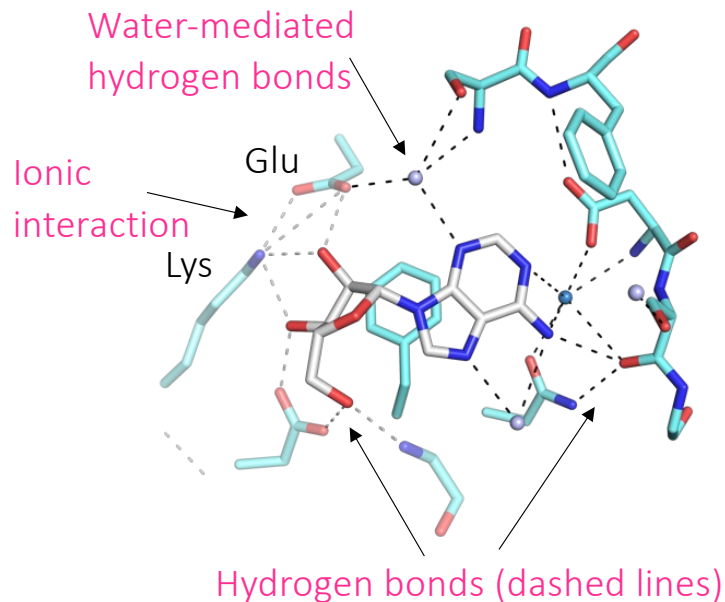
- Two polar groups interact: $O-H \cdots O$; $N-H \cdots O$; $N-H \cdots N$; $O-H \cdots N$

• Ionic interactions

- Positively charged Arg or Lys (or His) with negatively charged Glu or Asp

3. Covalent interactions

- Disulphide bonds (formed between two cysteine residues)
- Metal ions e.g. Mg^{2+} , Mn^{2+} , Cu^{2+} , Zn^{2+}

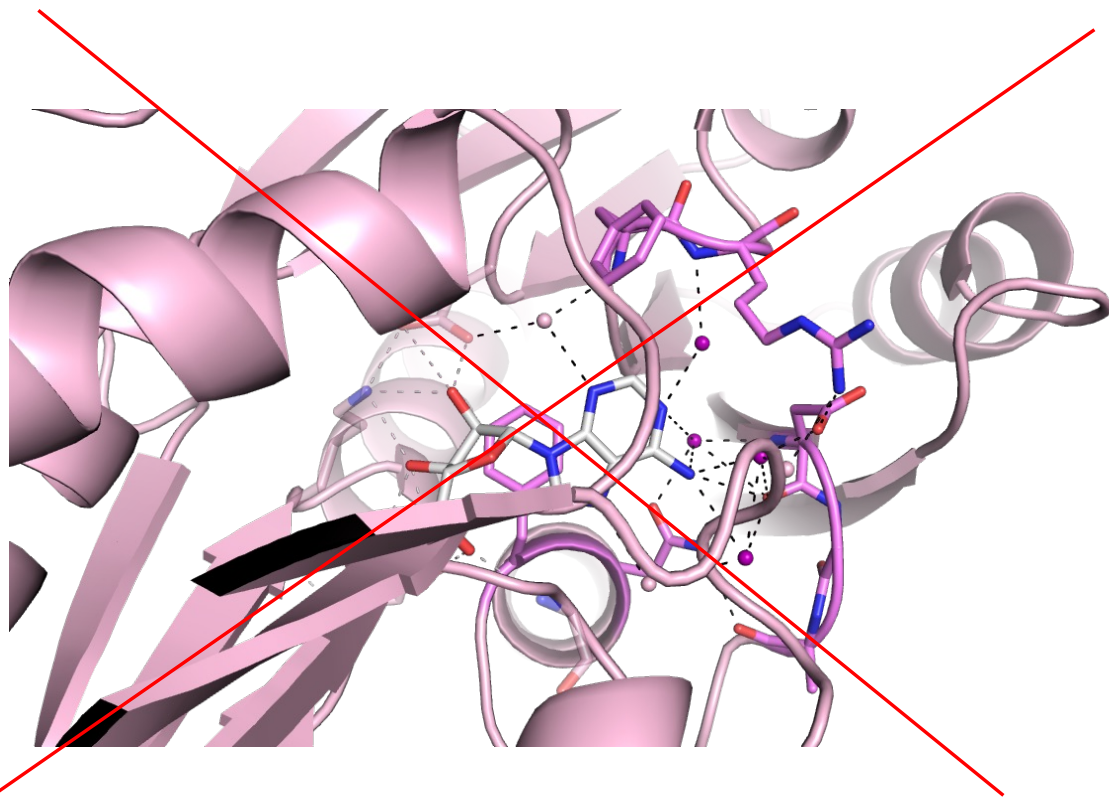


How to make a good protein figure:

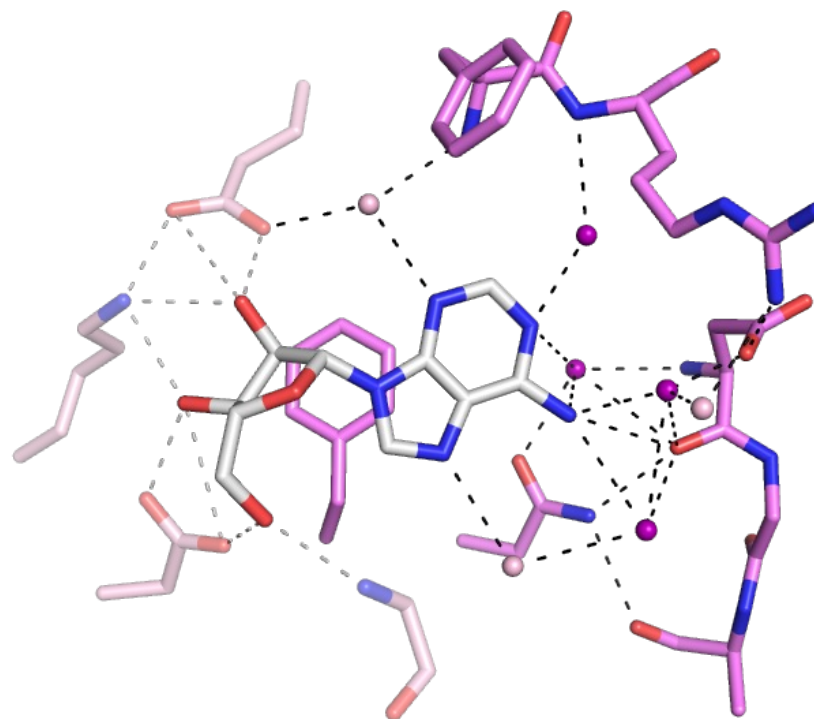
1. Make it **clear and simple!**
 - Highlight the most important things and **remove distractions**
2. Use colors, labels, arrows etc. wisely
 - Select **contrasting colors** to highlight differences
 - Label the most important things (e.g. catalytic residues, secondary structures involved in interactions)
3. Choose a style/styles that is suitable for what you want to show
 - E.g. cartoon, sticks and surface representations can be used to highlight different aspects of a protein
4. Write a **clear and descriptive figure legend!**
 - All colors, signs etc. should be clearly explained
 - PDB/AlphaFold structures used should be referenced

Example: Ligand-binding interactions

Focus on your message!
Visualization style and color



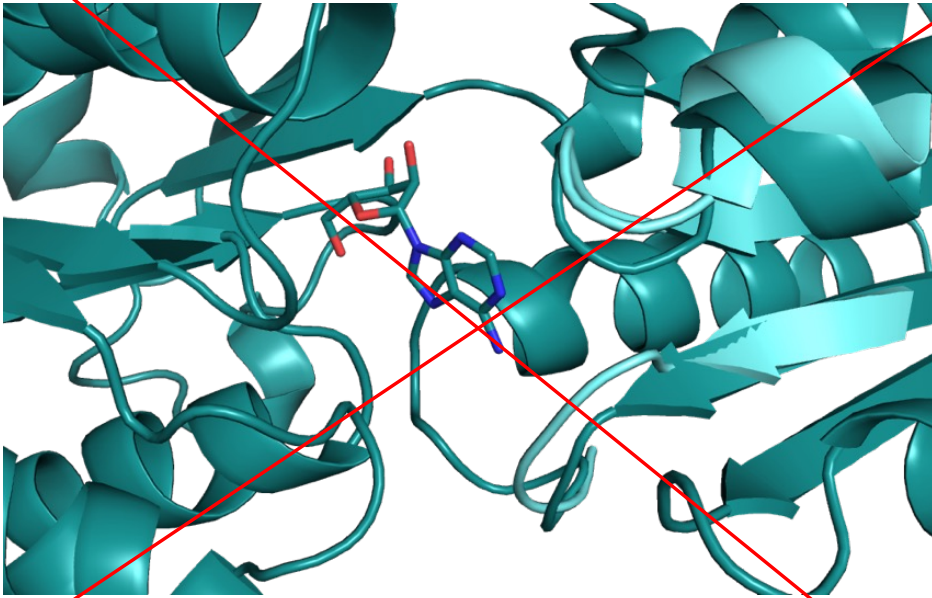
Location of the binding site



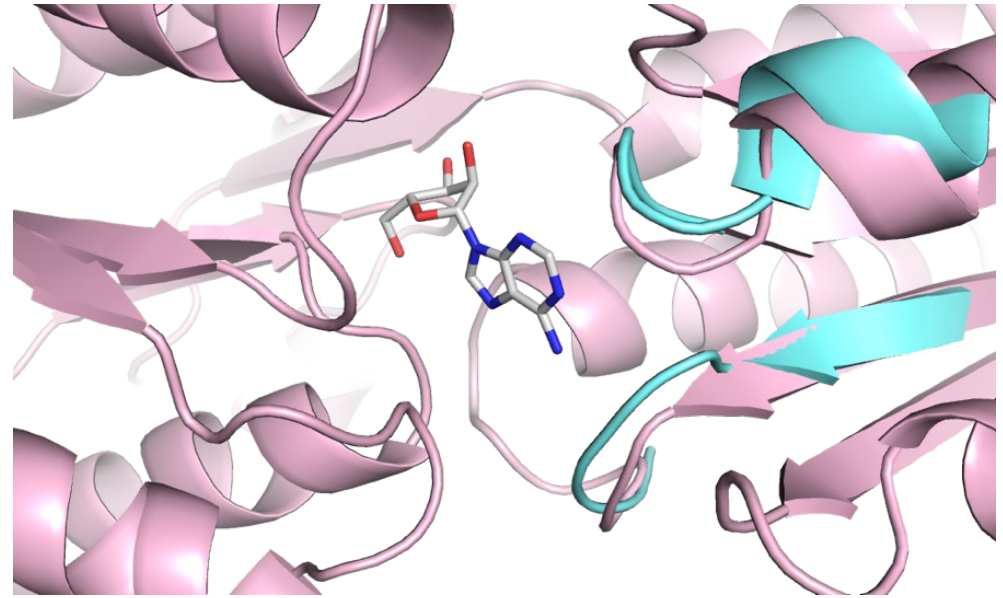
Interactions between ligand (white)
and protein (pink)

Example: Two superimposed structures

Colors do matter!



The two colors are too similar to show differences without and with ligand



The structure without ligand (cyan) and with ligand (pink) -> The differences are clearly visible!

Example: Figure and legend

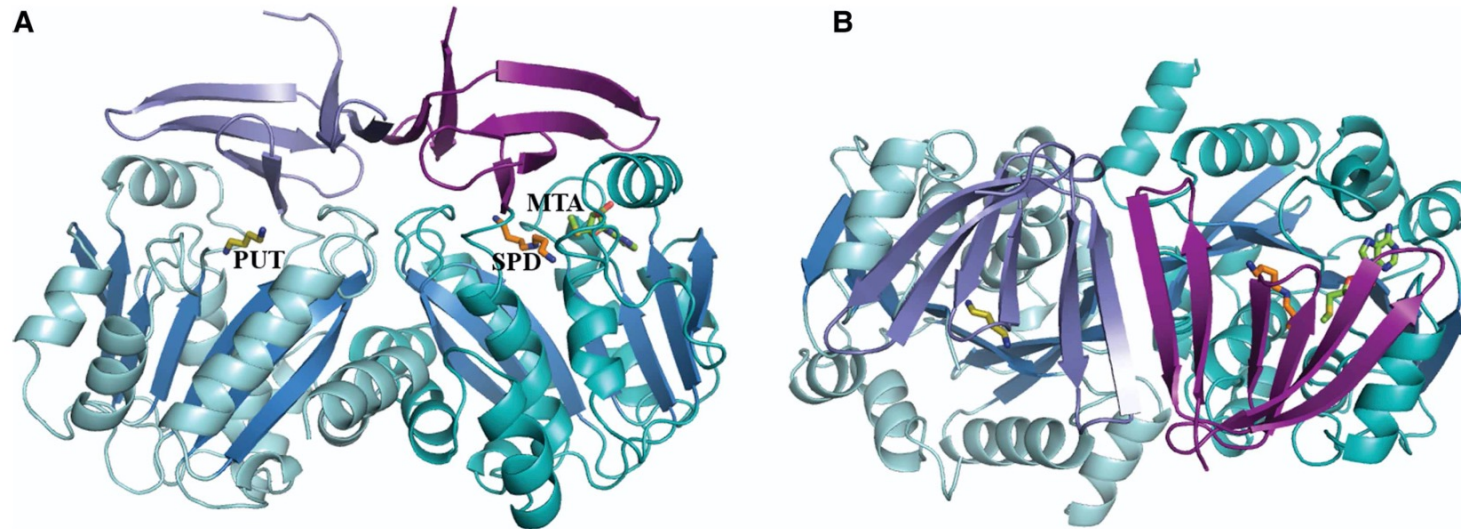


Figure 2. Structure of SySPDS with bound substrate and product.

Side (A) and top (B) view of the SPDS dimer with the N-terminal β -stranded domain of each monomer shown in light and dark purple, respectively and the C-terminal Rossmann fold-like domain with the β -stranded core (blue) shown in cyan and turquoise, respectively. Putrescine (PUT, olive), spermidine (SPD, orange) and 5'-methylthioadenosine (MTA, green) are shown in sticks.

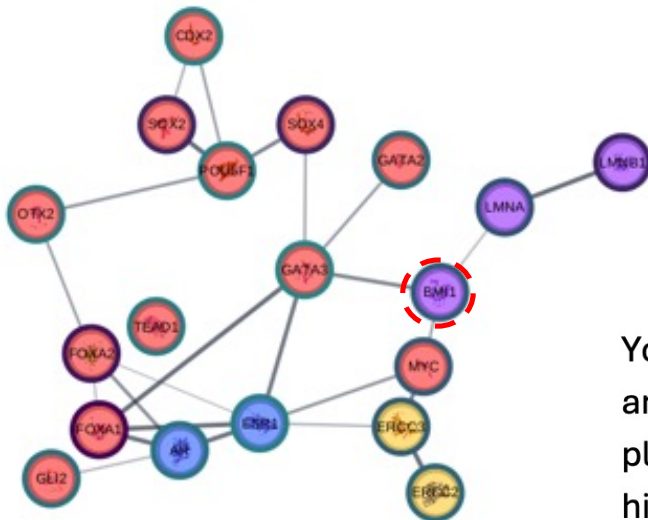
Title for the who figure

All Colors Explained

Text
Explain what is
shown and
how

Exercise – To get familiar with Pymol

Visualization of protein interactions



PyMOL exercise

Bmi1/Ring1b-UbcH5c complex structure

You will look at the 3D structures of three interacting proteins including RING2 (chain C) and BMI1 (chain B) both being part of human polycomb repressive complex 1-like, which plays a role in histone and gene regulation as describe by Wang et al. (2004) in “Role of histone H2A ubiquitination in polycomb silencing”. These two proteins interact as a heterodimer with the E2 enzyme UbcH5c (chain A) and the complex structure has been crystallized by Bentley et al. (2011) “Recognition of UbcH5c and the nucleosome by the BMI1/RING1b ubiquitin ligase complex” (PDB ID 3RPG). Now you will analyze how these proteins interact with each other.

3D structures from PDB

- Includes 3D structures from macromolecules
 - Proteins, DNA, RNA
- Established in 1971 at Brookhaven National Laboratory
 - Contained 7 structures!
 - Helped to store and distribute structural data
- Structures have been determined using
 - X-ray crystallography
 - Nuclear Magnetic Resonance
 - Electron microscopy (cryo-EM)



CRYSTALLOGRAPHY

Protein Data Bank

A repository system for protein crystallographic data will be operated jointly by the Crystallographic Data Centre, Cambridge, and the Brookhaven National Laboratory. The system will be responsible for storing atomic coordinates, structure factors and electron density maps and will make these data available on request. Distribution will be on magnetic tape in machine-readable form whenever possible. There

Crystallography: Protein Data Bank. *Nature New Biology* **233**, 223 (1971).
<https://doi.org/10.1038/newbio233223b0>

Link to the article in PDB

Recognition of Ubch5c and the nucleosome by the Bmi1/Ring1b ubiquitin ligase complex

Matthew L Bentley¹, Jacob E Corn¹,
Ken C Dong², Qui Phung³,
Tommy K Cheung³ and Andrea G Cochran^{1,*}

¹Department of Early Discovery Biochemistry, Genentech Research and Early Development, South San Francisco, CA, USA, ²Department of Structural Biology, Genentech Research and Early Development, South San Francisco, CA, USA and ³Department of Protein Chemistry, Genentech Research and Early Development, South San Francisco, CA, USA

The Polycomb repressive complex 1 (PRC1) mediates gene silencing, in part by monoubiquitination of histone H2A on lysine 119 (uH2A). Bmi1 and Ring1b are critical components of PRC1 that heterodimerize via their N-terminal RING domains to form an active E3 ubiquitin ligase. We have determined the crystal structure of a complex between the Bmi1/Ring1b RING–RING heterodimer and the E2 enzyme Ubch5c and find that Ubch5c interacts with Ring1b only, in a manner fairly typical of E2–E3 interactions. However, we further show that the Bmi1/Ring1b RING domains bind directly to duplex DNA through a basic surface patch unique to the Bmi1/Ring1b RING–RING dimer. Mutation of residues on this interaction surface leads to a loss of H2A ubiquitination activity. Computational modelling of the interface between Bmi1/Ring1b–Ubch5c and the nucleosome suggests that Bmi1/Ring1b interacts with both nucleosomal DNA and an acidic patch on histone H4 to achieve specific monoubiquitination of H2A. Our results point to a novel mechanism of substrate recognition, and control of product formation, by Bmi1/Ring1b.

PDB Search for the structure

RCSB PDB Deposit Search Visualize Analyze Download Learn About Documentation Careers COVID-19 MyPDB Contact us

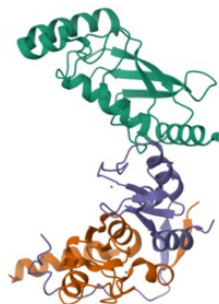
RCSB PDB PROTEIN DATA BANK 223,166 Structures from the PDB 1,068,577 Computed Structure Models (CSM)

3D Structures 3RPG Include CSM Advanced Search Browse Annotations Help

PDB-101 wwPDB EMDatResource NAKB wwPDB Foundation PDB-Dev

Structure Summary Structure Annotations Experiment Sequence Genome Versions

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (ZN)

Global Symmetry: Asymmetric - C1
Global Stoichiometry: Hetero 3-mer - A1B1C1

3RPG

Bmi1/Ring1b-Ubch5c complex structure
PDB DOI: <https://doi.org/10.2210/pdb3RPG/pdb>

Classification: **LIGASE**
Organism(s): *Homo sapiens*
Expression System: *Escherichia coli*
Mutation(s): No

Deposited: 2011-04-26 Released: 2011-08-17
Deposition Author(s): Bentley, M.L., Dong, K.C., Cochran, A.G.

Experimental Data Snapshot
Method: X-RAY DIFFRACTION
Resolution: 2.65 Å
R-Value Free: 0.243
R-Value Work: 0.217
R-Value Observed: 0.219
Starting Models: experimental
[View more details](#)

wwPDB Validation 3D Report Full Report

Metric	Percentile Ranks	Value
Rfree		0.237
Clashscore		11
Ramachandran outliers		0
Sidechain outliers		0.6%
RSRZ outliers		0.9%

Worse Better
■ Percentile relative to all X-ray structures
■ Percentile relative to X-ray structures of similar resolution

PDB information on the 3 proteins in the 3PRG complex structure

Macromolecules

Find similar proteins by: Sequence (by identity cutoff) | 3D Structure

Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details	Image
Ubiquitin-conjugating enzyme E2 D3	A	149	Homo sapiens	Mutation(s): 0 Gene Names: UBE2D3 , UBC5C , UBCH5C EC: 6.3.2.19	

UniProt & NIH Common Fund Data Resources

Find proteins for [P61077](#) (*Homo sapiens*)Explore [P61077](#) Go to UniProtKB: [P61077](#)

Entity ID: 2

Molecule	Chains	Sequence Length	Organism	Details	Image
Polycomb complex protein BMI-1	B	117	Homo sapiens	Mutation(s): 0 Gene Names: BMI1 , PCGF4 , RNF51	

UniProt & NIH Common Fund Data Resources

Find proteins for [P35226](#) (*Homo sapiens*)Explore [P35226](#) Go to UniProtKB: [P35226](#)

Entity ID: 3

Molecule	Chains	Sequence Length	Organism	Details	Image
E3 ubiquitin-protein ligase RING2	C	121	Homo sapiens	Mutation(s): 0 Gene Names: RNF2 , BAP1 , DING1 , HIPI3 , RING1B EC: 6.3.2	

UniProt & NIH Common Fund Data Resources

Find proteins for [Q99496](#) (*Homo sapiens*)Explore [Q99496](#) Go to UniProtKB: [Q99496](#)

Biological assembly 1 assigned by authors.

Macromolecule Content

- Total Structure Weight: 44.26 kDa
- Atom Count: 2,912
- Modelled Residue Count: 349
- Deposited Residue Count: 387
- Unique protein chains: 3

Literature

[Download Primary Citation](#)

Recognition of Ubch5c and the nucleosome by the Bmi1/Ring1b ubiquitin ligase complex.

[Bentley, M.L., Corn, J.E., Dong, K.C., Phung, Q., Cheung, T.K., Cochran, A.G.](#)
(2011) EMBO J 30: 3285-3297

PubMed: [21772249](#) [Search on PubMed](#) [Search on PubMed Central](#)

DOI: <https://doi.org/10.1038/emboj.2011.243>

Primary Citation of Related Structures:

3RPG

PubMed Abstract:

The Polycomb repressive complex 1 (PRC1) mediates gene silencing, in part by monoubiquitination of histone H2A on lysine 119 (uH2A). Bmi1 and Ring1b are critical components of PRC1 that heterodimerize via their N-terminal RING domains to form an active E3 ubiquitin ligase. We have determined the crystal structure of a complex...

[View More](#)

Organizational Affiliation:

Department of Early Discovery Biochemistry, Genentech Research and Early Development, South San Francisco,

Small Molecules

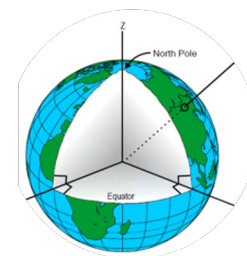
Ligands (1 Unique)

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
ZN	D [auth B], E [auth B], F [auth C], G [auth C]	ZINC ION Zn PTFCDOLFPIGGS-UHFFFAOYSA-N		Interactions Interactions & Density
Download Ideal Coordinates CCD File Download Instance Coordinates				

Chain A = Ubiquitin-conjugating enzyme E2 D3
Chain B = Polycomb complex protein BMI-1
Chain C = E3 ubiquitin-protein ligase RING2

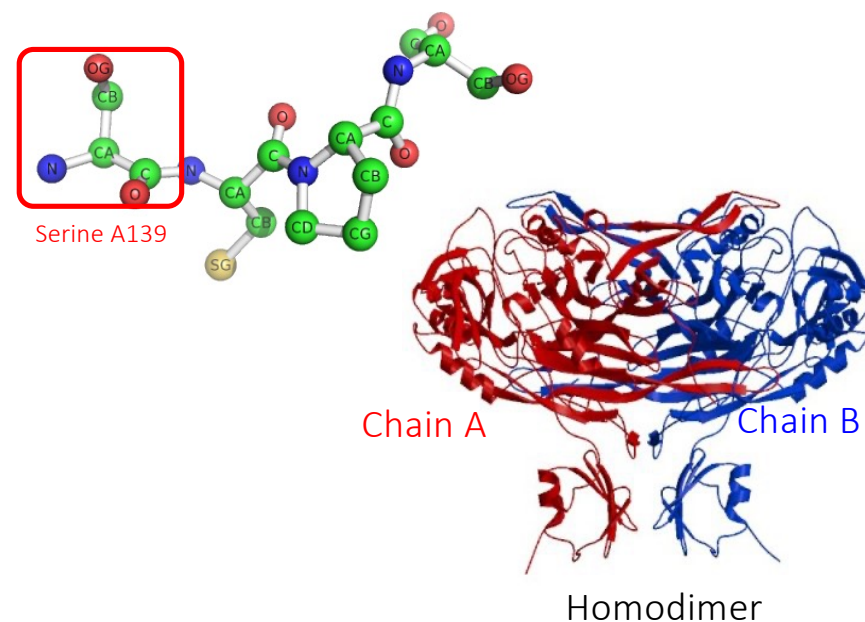
The PDB File

Atomic coordinates describe the location in space for each atom in the protein



Atom	Chain identifier		Residue number	X	Y	Z	Occupancy	B-factor
	Amino acid							
ATOM	1	N	SER A 139	22.479	-4.111	18.945	1.00	34.34
ATOM	2	CA	SER A 139	22.123	-2.660	18.772	1.00	36.15
ATOM	3	C	SER A 139	21.917	-2.021	20.148	1.00	35.23
ATOM	4	O	SER A 139	21.829	-2.724	21.148	1.00	37.66
ATOM	5	CB	SER A 139	20.891	-2.485	17.863	1.00	36.40
ATOM	6	OG	SER A 139	19.742	-3.154	18.362	1.00	43.71
ATOM	7	N	CYS A 140	21.832	-0.699	20.188	1.00	34.49
ATOM	8	CA	CYS A 140	21.716	0.023	21.450	1.00	36.17
ATOM	9	C	CYS A 140	20.371	0.642	21.738	1.00	37.47
ATOM	10	O	CYS A 140	19.731	1.189	20.844	1.00	36.62
ATOM	11	CB	CYS A 140	22.749	1.139	21.490	1.00	35.87
ATOM	12	SG	CYS A 140	24.467	0.525	21.122	1.00	36.23
ATOM	13	N	PRO A 141	20.006	0.713	23.028	1.00	38.50
ATOM	14	CA	PRO A 141	18.735	1.279	23.481	1.00	38.39
ATOM	15	C	PRO A 141	18.517	2.743	23.086	1.00	39.20
ATOM	16	O	PRO A 141	19.407	3.590	23.232	1.00	38.41
ATOM	17	CB	PRO A 141	18.809	1.093	25.001	1.00	39.58
ATOM	18	CG	PRO A 141	20.284	1.177	25.286	1.00	38.94
ATOM	19	CD	PRO A 141	20.849	0.333	24.181	1.00	39.16
ATOM	20	N	SER A 142	17.342	3.006	22.515	1.00	37.85
ATOM	21	CA	SER A 142	16.939	4.352	22.097	1.00	39.22
ATOM	22	C	SER A 142	17.444	4.856	20.744	1.00	34.22
ATOM	23	O	SER A 142	17.045	5.945	20.316	1.00	31.72
ATOM	24	CB	SER A 142	17.288	5.385	23.185	1.00	40.67
ATOM	25	OG	SER A 142	16.498	5.192	24.346	1.00	48.68

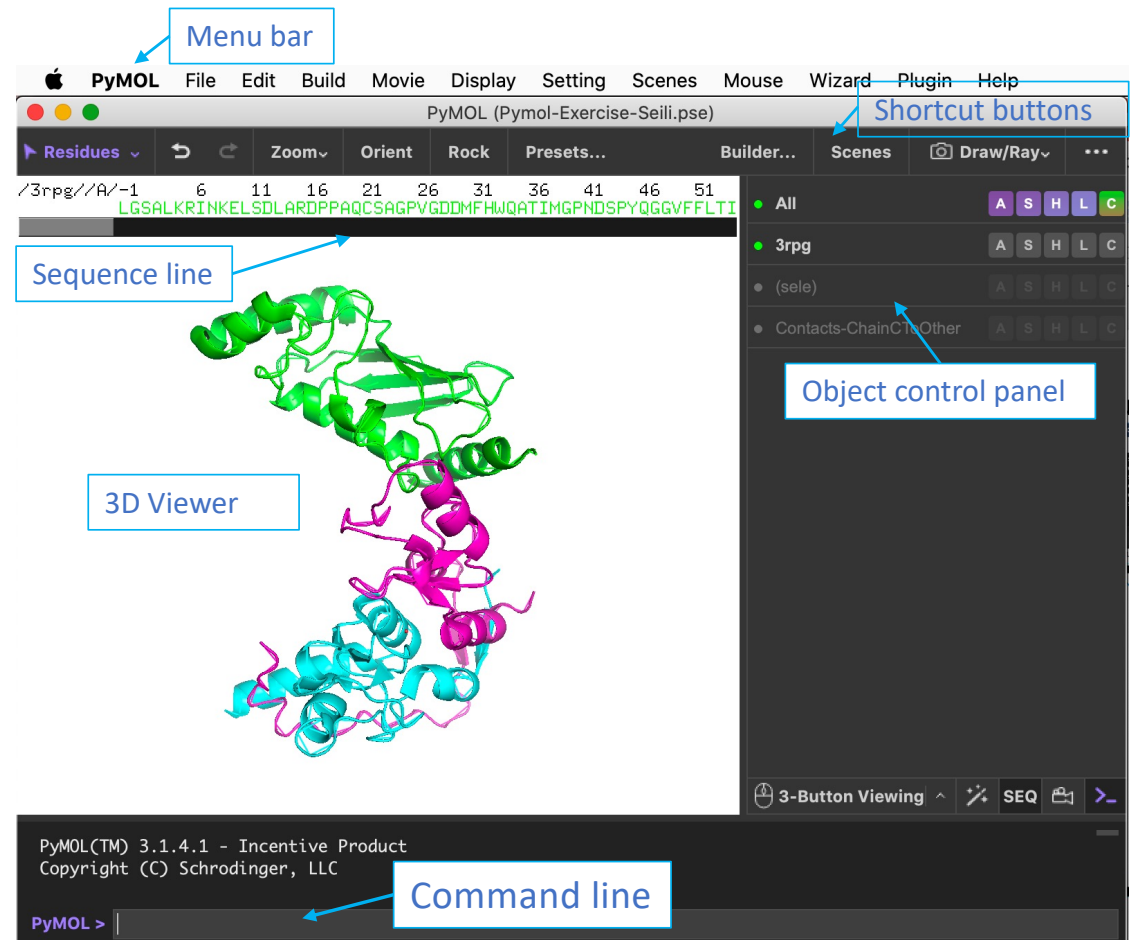
N
C
C
O
C
C
O
S
N
C
C
C
C
N
C
C
O
C



PDB Identification Code (PDB ID), unique for each structure, is a 4-character alphanumeric identifier e.g. 1AOX

PyMOL – The program we will use!

- Created by Warren DeLano
 - First release in 2000
 - Python script, OpenGL, open source
 - Since 2010 supported and distributed by Schrödinger, Inc.
- Possibility to save sessions – Do it!
- High quality rendering, movies etc.
- Automatic retrieval of molecules from PDB
- Possibility to use scripts to do complicated tasks
- Allow user to fully control and adjust the items displayed



<https://en.wikipedia.org/wiki/PyMOL>

Pymol exercise and Group work

1. Individual Pymol exercise Everyone in the group will do the exercise to learn how to use Pymol
 - Have you installed PyMol on your computer?
 - Save the license file from GitHub
 - Open Pymol and read the license file in Pymol
 - Follow the detailed information in the Pymol exercise instructions (GitHub)

2. Group work :

- Work with the AlphaFold models

Project presentation and report

1. Analyze the complex structure with Pymol
2. Get familiar with the proteins using literature, UniProt and PDB – Please, remember to provide the source citation!
3. Design what kind of figures and figure legends you will prepare *to explain how the protein complex functions*
4. Submit figures and figure legends in the Project presentation and report

QUALITY ASSESSMENT OF 3D MODELS

1. DOES THE 3D MODEL AGREE WITH BIOLOGICAL DATA?

- *STUDY UniProt Structure Section & biological data and PDB*
- *READ Literature*
- *ANALYZE your own functional predictions and experimental results*
- 2. MODEL VALIDATION (if experimental structure is available):
 - *COMPARE the 3D model with experimental PDB structures of the protein*
- 3. FOR PUBLICATION:
 - *ASSESS the quality of the 3D model with special programs e.g. MolProbity, QMEAN4 and MODFOLD9*

Practical tips

For 3D complex analysis:

- Locate the interacting parts (UniProt: Family & Domains, AlphaFold: PAE map)
- Find out there are experimental complex structures available (UniProt Structure)
- In Pymol, hide the unstructured regions and create a new object in Pymol
- Use Pymol to identify the interaction regions and study if there are structures available for those regions (UniProt)
- In UniProt: Find functional details, known mutations (UniProt Features) that affect the specific interaction and analyze where the mutation is in the 3D model. Does it have any effect on protein-protein interactions?



**Questions are
welcome &
Ask for help when
you need it!**